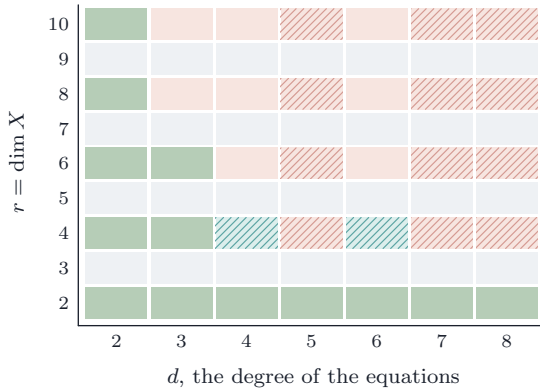




-  proved for every N : the theorem of Lefschetz for $r = 2$, the very general theorem for $d = 2$, and for $d = 3$, $r \leq 6$
-  proved given the analogue of Achter–Pries for cyclic covers of degree 4 and 6, by Markman’s theorem
-  open: Weil classes of abelian r -folds over $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-3})$, from dimension 6 or 8 on
-  open for N large: Weil classes of $\mathbb{Q}(\zeta_e)$, $\varphi(e) \geq 4$, the case $P2_{\text{split}}$ of the paper
-  r odd: no Hodge classes of degree r , nothing to prove